

Chrono Labeler

User Manual

User Interface:

As a minimalist device, the Chrono Labeler (from now on "the device") has only a button and 3 LEDs.

Blue LED: WiFi indicator, which has 4 possible states:

- OFF: WiFi is off and you have a nice (but very light) paperweight.
- Flashing: Access Point mode. The device is ready for initial configuration via its WEB page. In this mode you can connect from your smartphone to its SSID ("Chrono Labeler") using the default password which, unless you change it in the source code, is "Blopa1961" (no quotes, but case sensitive).
Then open Firefox or Chrome and go to 10.0.0.1 to access the main WEB page where you will find a link to the WiFi configuration.
- Blinking: the device is trying to connect to your Router.
- Steady ON: The device is connected to your Router.

Yellow LED: NTP indicator as follows:

- OFF: time is invalid because the device is not connected to the Router or the NTP server is unreachable.
- Blinking: the device is connected to the router and trying to obtain the current time from the Internet by NTP.
- Steady ON: time was acquired and we're ready to print (if the printer is paired and on).

Green LED: busy/printing

- OFF: the device is not communicating with a printer.
- Blinking: the device is trying to connect to the currently selected printer because of a print command. If the device cannot connect to the printer, the command will timeout after 12 seconds.
- Steady ON: the device is sending data to the printer.

Button:

Pressing and holding the button for more than 1.5 seconds changes the WiFi mode (AP mode, Connected to Router or OFF). If for some reason you can't access the device from the WEB page and want to change the configuration, press and hold the button until the blue LED changes to flashing, release the button and connect as explained in the blue LED description above.

Normally, when the device is online and has acquired the time it will print a sticker each time you press the button. If the device does not have a valid time (NTP server connection failed or WiFi is off) the button will be ignored. Both the blue and yellow LEDs must be steady on to print dated stickers. If for whatever reason (off, unpaired) the printer cannot be reached within 12 seconds, the command will be ignored (no errors, no blink codes, nothing, the green LED just stops blinking).

WEB interface:

The first thing you need to do is connect the device to your WiFi (router).

- Make sure the blue LED is flashing

- From your smartphone connect to the SSID "Chrono Labeler" using the password you defined in MYPASS in the source code (default is Blopa1961).
- Open Firefox or Chrome in you smartphone and visit 10.0.0.1
- Click the "Configure WiFi" link there
- SSID: Here you can type your router's SSID or press the "Scan WiFi Networks" button, wait a few seconds and choose your router from the listed links.
- Password: Type your router's password (NOT this device's)
- IP: here you can type a fixed IP (granted that you know your local network configuration and free IPs) or you can leave it blank for DHCP mode (your router will automatically assign an IP).
- Gateway and Mask: if you typed a fixed IP, type the corresponding values here. If you chose DHCP mode leave them blank.
- Retry: when for some reason the device gets disconnected from your router, it will try to reconnect for this many minutes (0 meaning forever). Once this timeout has elapsed unsuccessfully, the device will switch back to AP mode.
- WiFi Startup Mode: you should choose "Connected to Router" here so the device connects as soon as you plug it in.
- Press SAVE. If all is correct, this configuration page will become invalid and the device will connect to your router (the blue LED should remain ON after blinking a few seconds).
- If your WiFi connection was successful, the yellow LED will start blinking, and if the NTP server is correctly defined it will change to steady ON after a few seconds (current time acquired by NTP).
- After a successful connection to your router you can access the device from your PC or smartphone using Firefox or Chrome (maybe Safari too, untested) by its assigned IP or visiting <http://chronolabeler.local>

Profiles:

This device can handle up to 12 printer profiles. Each profile has unique name and all parameters necessary to describe a sticker (size, fonts, margins, printer name, etc.)

Profiles are defined in the "Configure Printer" WEB page. There are 3 predefined profiles, but none can be used until you PAIR your printer.

The selected profile can be modified (if you don't change the profile name), it will be updated when you press SAVE.

To pair a printer:

- Select a profile
- Edit it as you like
- Press the PAIR button
- Wait a few seconds
- A list of detected BLE printers will appear above the PAIR button
- You can press PAIR again if your printer is not listed (is it on? LOL)
- Select your desired printer by clicking it's name in the list
- Connection will change to "Paired"
- Press SAVE
- The device is ready to print and will go back to the main page
- If you clicked PAIR, did **NOT** select the printer from the BLE list and pressed SAVE, and a printer with the first part of the BLE name matching the text under "Bluetooth Name" is found, it will be paired and saved. You will see its full BLE name and "paired" if you go back to the Printer Configuration page.

To create a new profile, select one with the same printer type (TP-110 or TP-220) and change (edit) the profile name. Please notice that if a profile with the exact same name you type here exists it will be overwritten when you press SAVE. Also, if there are already 12 profiles defined; the selected profile will be overwritten without confirmation when you SAVE it, regardless of if the profile name matches or not.

More than one profile with different sticker sizes, fonts, etc. can be defined for the same printer.

Pressing DELETE will delete the selected profile. At least 1 profile must always be active, so if all profiles are deleted, the 3 default profiles will be reloaded (and you will have to PAIR a printer again).

The last printer you SAVE in this page will become the DEFAULT printer, that is, the printer to where the sticker will be sent when you press <THE BUTTON>. You can SAVE an unmodified profile (even with the printer off) to make it the default. Please notice that if you chose a different printer from the main page's pull-down, it will remain active until you change it or the device is restarted or power cycled.

Date and Time configuration:

In this page you can choose whichever format and separators you like for your Date and Time, be it MM/DD/YY, DD/MM, 24 hour, 12 hour AM/PM and whether the day of the week gets printed or not. You can use PREVIEW as many times as you like while you change the formats. The preview interface is not only WYSIWYG, it's an actual preview of the buffer that will be sent to the printer.

Main Page:

Here you can select a printer from the profiles pull-down. When you change printers the status will change to disconnected (or **Not Paired** if the printer in the selected profile isn't paired).

Once a sticker has been printed the connection will remain active unless you change or turn off the selected printer (or it goes to standby), which means that if you print another sticker it will be faster.

If you type some text in the "Text to print" fields, you can quickly print a label in the selected font(s).

Pressing the PRINT button in this WEB page when the first "Text to print" line is blank will output a Date and Time sticker just as if you had pressed <THE BUTTON>.

The print button will be grayed out if the selected printer is not paired.